# IGCSE 0607 Practice Questions — E9.2 — Combined Questions & Markscheme

## Relative and expected frequencies

**Total Questions:** 15 — **Total Marks:** 104 Marks

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**1** A biased five-sided spinner is numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number.

| **Number** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Probability** | 0.15 | 0.25 | $x$ | 0.2 | $3x$ |

**(a)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** The spinner is spun 250 times.
Calculate the expected number of times the spinner lands on a number greater than 3.

..... **[2]**

**2** A spinner has five sections, labelled $A$, $B$, $C$, $D$ and $E$.
The table shows the relative frequencies of the spinner landing on each section after 200 spins.

| Section | $A$ | $B$ | $C$ | $D$ | $E$ |
|---|---|---|---|---|---|
| Relative frequency | 0.15 | 0.3 | $x$ | 0.25 | 0.1 |

**(a)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** Estimate the number of times the spinner lands on section $B$ in 200 spins.

..... **[2]**

**3** A biased 4-sided spinner, numbered 1, 2, 3, and 4, is spun 200 times.
The table shows the frequencies of each outcome.

| Number | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Frequency | 45 | 35 | 70 | 50 |

**(a)** Write down the relative frequency of the spinner landing on 3.

..... **[1]**

**(b)** The spinner is spun 1000 times.
Estimate the number of times it will land on 1.

..... **[2]**

**(c)** Is the spinner fair?
Give a reason for your answer.

..... **[1]**

**(d)** Explain why spinning the spinner 1000 times gives a more reliable estimate of the probability than spinning it 200 times.

..... **[1]**

**4** A biased 5-sided dice has faces numbered 1, 2, 3, 4, and 5.
Maya rolls the dice 200 times and records the frequency of each score in the table below.

| Score on dice | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Frequency | 32 | 28 | 54 | 46 | 40 |

**(a)** Write down the relative frequency of rolling a score of 3, giving your answer as a decimal.

..... **[1]**

**(b)** Explain why the results in the table suggest that the dice is biased.

..... **[1]**

**(c)** Maya rolls the dice a further 500 times.
Calculate an estimate for the number of times she will roll an odd score.

..... **[3]**

**(d)** Maya rolls the dice $n$ times and expects to roll a score of 5 exactly 36 times.
Find the value of $n$.

$n =$ ..... **[2]**

**5** A biased four-sided die is rolled 60 times.
The scores are 1, 2, 3, and 4.
The frequencies of each score are shown in the table.

| Score | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Frequency | $x$ | $2x + 5$ | $3x - 2$ | $x + 8$ |

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Find the relative frequency of rolling a 3.

..... **[1]**

**(c)** The die is rolled 1200 times.
Calculate the expected number of times it rolls a 2.

..... **[2]**

**6** A biased spinner has three sectors labelled $A$, $B$, and $C$.
In an experiment, the spinner is spun $(x^2 + 10x)$ times, where $x > 0$.
Sector $A$ occurs $(2x + 4)$ times and sector $B$ occurs $(5x + 2)$ times.
The relative frequency of the spinner landing on sector $A$ is $\frac{1}{6}$.

**(a)** Show that $x^2 - 2x - 24 = 0$.

**[2]**

**(b)** Solve the equation $x^2 - 2x - 24 = 0$ and hence find the total number of times the spinner was spun.

Total spins = ..... **[3]**

**(c)** Using the results of this experiment, find, as a fraction in its simplest form, the relative frequency of the spinner landing on

**(i)** sector $B$,

..... **[2]**

**(ii)** sector $C$.

..... **[2]**

**(d)** In a game, a player spins the spinner 600 times.
The player wins \$6 each time the spinner lands on sector $A$, wins \$2 each time it lands on sector $B$, and loses \$3 each time it lands on sector $C$. Calculate the player's expected net profit

after 600 spins.

\$ ..... **[3]**

**7** A spinner has four colours: Red, Blue, Green, and Yellow.
A student spins it 50 times and records the results.
The table shows the relative frequencies for three of the colours.

| Colour | Red | Blue | Green | Yellow |
|---|---|---|---|---|
| Relative frequency | 0.15 | 0.45 | 0.10 | $x$ |

**(a)** Find the value of $x$.

$x =$ ..... **[1]**

**(b)** The spinner is spun 300 times in total.
Estimate the number of times it lands on Blue.

..... **[2]**

**(c)** The spinner is meant to be fair.
Comment on whether the results show it is fair or biased.

..... **[1]**

**8** A biased coin is tossed $n$ times.
The probability of getting a head on each toss is $p$.
The expected number of heads is 30 and the expected number of tails is 45.

**(a)** Find the value of $n$.

$n =$ ..... **[1]**

**(b)** Calculate the value of $p$.

$p =$ ..... **[2]**

**(c)** The same coin is now tossed 300 times.
Work out the expected number of tails.

..... **[3]**

**9** A bag contains a large number of coloured beads.
A bead is chosen at random, its colour recorded, and then returned to the bag.
This experiment is repeated 300 times.
The table shows the frequencies of the colours drawn.

| Colour | Red | Blue | Green | Yellow |
|---|---|---|---|---|
| Frequency | 84 | 105 | $x$ | $y$ |

**(a)** The relative frequency of choosing a Green bead is 0.22.
Work out the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** There are exactly 15 Yellow beads in the bag.
Estimate the total number of beads in the bag.

..... **[2]**

**(c)** A new experiment is done where a bead is chosen and replaced 800 times.
Calculate the expected number of times a Blue or Green bead is chosen.

..... **[2]**

**10** A researcher records the colours of cars passing a checkpoint.
Over one week, 800 cars pass the checkpoint.
The frequencies of the colours are shown in the table.

| **Colour** | Silver | Black | White | Other |
|---|---|---|---|---|
| **Frequency** | 260 | 180 | $w$ | $w + 40$ |

**(a)** Find the relative frequency of a car being Silver.
Give your answer as a fraction in its simplest form.

..... **[2]**

**(b)** Find the value of $w$.

$w =$ ..... **[2]**

**(c)** The researcher uses these results to estimate the probability that the next car passing the checkpoint is White.
Work out this estimate.

..... **[1]**

**(d)** In one month, 12000 cars are expected to pass the checkpoint.
Calculate the expected number of Black cars.

..... **[2]**

**11** A quality control inspector tests a random sample of 250 light bulbs produced by a factory.
The table shows the lifetimes, $t$ hours, of these 250 light bulbs.
The relative frequency of a light bulb having a lifetime of more than 1500 hours is 0.42.

| Lifetime ($t$ hours) | $0 < t \leq 500$ | $500 < t \leq 1000$ | $1000 < t \leq 1500$ | $1500 < t \leq 2000$ | $2000 < t \leq 2500$ |
|---|---|---|---|---|---|
| Frequency | 15 | $x$ | 95 | $y$ | 35 |

**(a)** Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** A batch of 4800 light bulbs is produced by the factory.
Calculate an estimate for the number of light bulbs in this batch that have a lifetime of 1000 hours or less.

..... **[3]**

**(c)** Explain why it is important that the sample of 250 light bulbs is chosen at random.

..... **[1]**

**(d)** The inspector tests a second random sample of 750 light bulbs from the same factory, and finds that 115 of them have a lifetime of more than 2000 hours.
Combine the results of both samples to find the best estimate for the probability that a light bulb has a lifetime of more than 2000 hours.

..... **[2]**

**12** A factory produces light bulbs.
The probability that a light bulb is defective is 0.025.
A sample of $N$ light bulbs is taken.
The expected number of defective light bulbs in this sample is 40.

**(a)** Calculate the value of $N$.

$N =$ ..... **[2]**

**(b)** A new quality control process is introduced.
The factory takes a new sample of 2400 light bulbs.
The expected number of non-defective light bulbs in this new sample is 2364.
Calculate the percentage decrease in the probability of a light bulb being defective.

..... % **[5]**

**13** A biased 5-sided spinner has sides numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number in terms of $x$.

| Number | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Probability | 0.15 | $2x$ | $x + 0.05$ | 0.2 | $x$ |

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** The spinner is spun 600 times.
Work out the expected number of times it lands on a prime number.

..... **[2]**

**(c)** A different spinner has 3 sections: Red, Blue, and Green.
The probability of landing on Red is $p$.
The probability of landing on Blue is $2p$.
The probability of landing on Green is $1 - 3p$.
This spinner is spun $N$ times.

The expected number of times it lands on Red is 54 and the expected number of times it lands on Green is 36.
Work out the value of $N$ and the value of $p$.

$N =$ ..... , $p =$ ..... **[4]**

**14** A biased four-sided spinner is numbered 1, 2, 3 and 4.
The table shows the probability of scoring each number.
The spinner is spun 400 times.
Let $E_1$ be the expected number of times the spinner lands on 1.
Let $E_3$ be the expected number of times the spinner lands on 3.
You are given that $E_1 \times E_3 = 5600$.
Also, the probability of scoring a 3 is greater than the probability of scoring a 1.

| **Score** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | $x$ | $2x$ | $y$ | $y$ |

**(a)** Show that $300x^2 - 100x + 7 = 0$.

**[3]**

**(b)** Solve the equation to find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(c)** Calculate the expected number of times the spinner lands on an even number.

..... **[2]**

**15** Two biased coins, Coin A and Coin B, are tossed.
The probability of Coin A landing on Heads is $p$.
The probability of Coin B landing on Heads is $p + 0.15$.
Coin A is tossed 300 times and Coin B is tossed 200 times.
The total expected number of Heads from these 500 tosses is 255.

**(a)** Form an equation in $p$ and show that $p = 0.45$.

**[3]**

**(b)** Both coins are tossed together 1000 times.
Work out the expected number of times they land on different outcomes.

..... **[3]**

**(c)** In a new experiment, Coin A is tossed $N$ times.
The relative frequency of landing on Heads is 0.48 and it lands on Heads 72 times.

**(i)** Work out the value of $N$.

$N =$ ..... **[2]**

**(ii)** Calculate the expected number of Tails in these $N$ tosses, using the theoretical probability $p = 0.45$.

..... **[2]**

**1** **(a)** $0.15 + 0.25 + x + 0.2 + 3x = 1$ **[M1]**

$x = 0.1$ **[A1]**

**(b)** $P(> 3) = P(4) + P(5) = 0.2 + 3(0.1) = 0.5$ **[M1]**

Expected number $= 250 \times 0.5 = 125$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**2** **(a)** $1 - (0.15 + 0.3 + 0.25 + 0.1)$ or $1 - 0.8$ **[M1]**

$0.2$ **[A1]**

**(b)** $200 \times 0.3$ **[M1]**

$60$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**3** **(a)** $\frac{70}{200}$ or $\frac{7}{20}$ or $0.35$ **[B1]**

**(b)** $1000 \times \frac{45}{200}$ **[M1]**

$225$ **[A1]**

**(c)** No, because the relative frequencies are not all close to 0.25 (or 50 out of 200) **[B1]**

**(d)** A larger number of trials leads to a relative frequency that is closer to the theoretical probability **[B1]**

[Back to Question] *[Total Marks: 5]*

**4** **(a)** $0.27$ **[B1]**

**(b)** If the dice were fair, each score would be expected to occur 40 times (or with relative frequency 0.20), whereas the observed frequencies differ considerably from 40 **[B1]**

**(c)** Total frequency of odd scores $= 32 + 54 + 40 = 126$ **[M1]**

Relative frequency of an odd score $= \frac{126}{200} = 0.63$ **[M1]**

$0.63 \times 500 = 315$ **[A1]**

**(d)** $\frac{40}{200} \times n = 36$ (or $0.2n = 36$) **[M1]**

$180$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**5** **(a)** $x + 2x + 5 + 3x - 2 + x + 8 = 60$ **[M1]**

$7x + 11 = 60 \implies 7x = 49$

$x = 7$ **[A1]**

**(b)** Frequency of $3 = 3(7) - 2 = 19$

$\frac{19}{60}$ **[B1]**

**(c)** Frequency of $2 = 2(7) + 5 = 19$

Relative frequency of $2 = \frac{19}{60}$

$1200 \times \frac{19}{60}$ **[M1]**

380 **[A1]**

[Back to Question] *[Total Marks: 5]*

**6 (a)** $\frac{2x+4}{x^2+10x} = \frac{1}{6}$ giving $x^2 + 10x = 6(2x + 4)$ **[M1]**

$x^2 + 10x = 12x + 24$ simplifying to $x^2 - 2x - 24 = 0$ **[A1]**

**(b)** $(x - 6)(x + 4) = 0$ **[M1]**

$x = 6$ (rejecting $x = -4$ since $x > 0$) **[A1]**

Total spins $= 6^2 + 10(6) = 96$ **[A1]**

**(c) (i)** Sector $B$ occurs $5(6) + 2 = 32$ times, giving $\frac{32}{96}$ **[M1]**

$\frac{1}{3}$ **[A1]**

**(ii)** Sector $C$ occurs $96 - 16 - 32 = 48$ times, giving $\frac{48}{96}$ **[M1]**

$\frac{1}{2}$ **[A1]**

**(d)** Expected number of spins: $A = 600 \times \frac{1}{6} = 100$, $B = 600 \times \frac{1}{3} = 200$, $C = 600 \times \frac{1}{2} = 300$ **[M1]**

Expected net profit $= (100 \times 6) + (200 \times 2) - (300 \times 3) = 600 + 400 - 900$ **[M1]**

100 **[A1]**

[Back to Question] *[Total Marks: 12]*

**7 (a)** $x = 1 - (0.15 + 0.45 + 0.10)$

$x = 0.30$ **[B1]**

**(b)** $300 \times 0.45$ **[M1]**

135 **[A1]**

**(c)** Biased, because the relative frequencies are not all close to 0.25 (or specifically Blue is 0.45, which is much higher than 0.25) **[B1]**

[Back to Question] *[Total Marks: 4]*

**8 (a)** 75 **[B1]**

**(b)** $\frac{30}{75}$ or $\frac{30}{30+45}$ **[M1]**

$0.4$ or $\frac{2}{5}$ **[A1]**

**(c)** $P(\text{tail}) = 1 - 0.4 = 0.6$ or $\frac{45}{75}$ **[B1]**

$300 \times 0.6$ **[M1]**

$180$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**9** **(a)** $\frac{x}{300} = 0.22 \implies x = 300 \times 0.22$ **[M1]**

$y = 300 - (84 + 105 + 66)$ **[M1]**

$x = 66$, $y = 45$ **[A1]**

**(b)** Relative frequency of Yellow $= \frac{45}{300}$ or $0.15$ **[M1]**

Total beads $= \frac{15}{0.15} = 100$ **[A1]**

**(c)** Probability of Blue or Green $= \frac{105+66}{300} = \frac{171}{300}$ or $0.57$ **[M1]**

$800 \times 0.57 = 456$ **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**10** **(a)** $\frac{260}{800}$ **[M1]**

$\frac{13}{40}$ **[A1]**

**(b)** $260 + 180 + w + w + 40 = 800 \implies 480 + 2w = 800$ **[M1]**

$w = 160$ **[A1]**

**(c)** $\frac{160}{800}$ or $\frac{1}{5}$ or $0.2$ **[B1]**

**(d)** $P(\text{Black}) = \frac{180}{800} = \frac{9}{40}$ **[M1]**

Expected number $= 12000 \times \frac{9}{40} = 2700$ **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**11** **(a)** $\frac{y+35}{250} = 0.42$ giving $y + 35 = 105$ **[M1]**

$y = 70$ **[A1]**

$x = 250 - (15 + 95 + 70 + 35) = 35$ **[A1]**

**(b)** Number of light bulbs with $t \leq 1000$ in sample $= 15 + 35 = 50$ **[M1]**

Estimated number $= \frac{50}{250} \times 4800$ **[M1]**

960 **[A1]**

**(c)** To avoid bias so that the sample is representative of all light bulbs produced by the factory **[B1]**

**(d)** $\frac{35+115}{250+750} = \frac{150}{1000}$ **[M1]**

0.15 (or $\frac{3}{20}$) **[A1]**

[Back to Question] *[Total Marks: 9]*

**12** **(a)** $0.025 \times N = 40$ or $N = \frac{40}{0.025}$ **[M1]**

1600 **[A1]**

**(b)** Expected defective $= 2400 - 2364 = 36$ **[B1]**

New probability of defective $= \frac{36}{2400} = 0.015$ **[B1]**

Decrease in probability $= 0.025 - 0.015 = 0.01$ **[M1]**

Percentage decrease $= \frac{0.01}{0.025} \times 100$ **[M1]**

40 **[A1]**

[Back to Question] *[Total Marks: 7]*

**13** **(a)** $0.15 + 2x + x + 0.05 + 0.2 + x = 1$ **[M1]**

$4x + 0.4 = 1 \implies 4x = 0.6 \implies x = 0.15$ **[A1]**

**(b)** $P(2,3,5) = 2(0.15) + (0.15 + 0.05) + 0.15 = 0.3 + 0.2 + 0.15 = 0.65$ **[M1]**

$600 \times 0.65 = 390$ **[A1]**

**(c)** Sets up both equations: $N \times p = 54$ and $N \times (1 - 3p) = 36$ **[M1]**

Expands the second equation: $N - 3Np = 36$ **[M1]**

Substitutes $Np = 54$ into the expanded equation: $N - 3(54) = 36$ **[M1]**

$N = 198$, $p = \frac{54}{198} = \frac{3}{11}$ (or 0.272...) **[A1]**

[Back to Question] *[Total Marks: 8]*

**14** **(a)** Sum of probabilities: $3x + 2y = 1 \implies y = \frac{1-3x}{2}$ **[B1]**

$E_1 = 400x$ and $E_3 = 400y \implies 400x \times 400y = 5600 \implies xy = \frac{7}{200}$ **[M1]**

Substitute for $y$: $x\left(\frac{1-3x}{2}\right) = \frac{7}{200} \implies 100x(1-3x) = 7 \implies 300x^2 - 100x + 7 = 0$

**[A1]**

**(b)** Factorising $(30x-7)(10x-1)=0$ or correct substitution into the quadratic formula **[M1]**

$x=\frac{7}{30}$ and $x=\frac{1}{10}$ (or 0.1) **[A1]**

When $x=\frac{7}{30}$, $y=0.15$ (reject as $y<x$). When $x=0.1$, $y=0.35$ (accept as $y>x$) **[M1]**

$x=0.1, y=0.35$ **[A1]**

**(c)** $P(\text{even})=2x+y=2(0.1)+0.35=0.55$ **[M1]**

Expected number $=400\times 0.55=220$ **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**15** **(a)** $300p+200(p+0.15)=255$ **[M1]**

$300p+200p+30=255$ **[M1]**

$500p=225 \implies p=\frac{225}{500}$

$p=0.45$ **[A1]**

**(b)** Probability Coin B is Heads $=0.45+0.15=0.60$

Probability of different outcomes = Probability (H, T) + Probability (T, H)

$(0.45\times 0.40)+(0.55\times 0.60)$ **[M1]**

$0.18+0.33=0.51$ **[M1]**

$1000\times 0.51=510$ **[A1]**

**(c)** **(i)** $\frac{72}{N}=0.48$ **[M1]**

$N=\frac{72}{0.48}$

$N=150$ **[A1]**

**(ii)** Probability of Tails $=1-0.45=0.55$

$150\times 0.55$ **[M1]**

82.5 **[A1]**

**[Back to Question]** *[Total Marks: 10]*